Principal Systems Engineer & Architect

ADDITIVE MANUFACTURING • CORE SYSTEMS (.NET / PYTHON) • DEVOPS & KERNEL SECURITY

Available for Custom Contracts

Battle-tested systems architect with deep, multi-disciplinary engineering expertise across additive manufacturing automation, production C# (.NET 8) daemons, Linux kernel sandboxing (Namespaces/Seccomp/AppArmor), and Android frameworks. Proven track record of architecting mission-critical infrastructure with zero compromise on memory management, defensive error traps, and multi-pass verification standards.

PRODUCTION PORTFOLIOS & FLAGSHIP PROJECTS

CORE COMPETENCIES

Additive & 3D Engineering

- Klipper
- OrcaSlicer
- STL Repair
- Marlin
- Input Shaping

Backend & Core Systems

- C# (.NET 8)
- Python 3.13
- FastAPI
- Node.js
- SQLite

DevOps & Security

- Debian / Linux
- Docker
- Seccomp BPF
- AppArmor
- CIS Baselines

Mobile & Automation

- Android ADB
- Zero-Root Debloat
- RF Sensing
- Systemd

Enterprise Freelance Automation Engine

`C# .NET 8` `SQLite` `Systemd` `Spectre.Console`

- **Browser Lock-Evasion:** Implemented transient GUID-tagged read-only SQLite cloning to query active browser session cookies without lock contention.
- **Automated ToS Protection:** Engineered real-time anti-circumvention NLP filters intercepting off-platform triggers to guarantee 100% platform policy compliance.
- **Headless Daemon & Telemetry:** Built a resource-clamped (<35MB RAM) background daemon polling every 30s with desktop alerts and PipeWire audio cues.

CONTRACT OFFERINGS

3D Mesh Repair & Slicer Tuning

Watertight STL repair, Klipper firmware calibration, and custom OrcaSlicer profiles.

Custom Software Prototyping

High-performance production utilities in C# (.NET 8) and Python with clean architectures.

Python Automation & Scraping

Headless browser scrapers, data extraction pipelines, and scheduled cron jobs.

Android Optimization & Debloat

Safe, non-root OEM telemetry and bloatware removal preserving Knox and Secure Folder.

DevOps Browser Cybersecurity Suite

`C# .NET 8` `Node.js (CDP)` `Linux LSM` `Puppeteer`

- **Host Sandbox & Kernel Auditing:** Automated validation of SUID root binaries (`4755`), unprivileged user namespaces, and Seccomp BPF syscall confinement.
- **Enterprise CIS Hardening:** Formulated and deployed enterprise policies for Firefox and Chrome enforcing DNS-over-HTTPS, telemetry blocks, and extension allowlists.
- **Dynamic Transport Security:** Orchestrated headless Chromium/Firefox via CDP to score CSP, HSTS preload, X-Content-Type-Options, and clickjacking mitigations.

Additive Manufacturing & Klipper Firmware Calibration

`PyVista` `Klipper / Marlin` `OrcaSlicer` `MeshFix`

- **Watertight STL Repair:** Algorithmic mesh repair pipelines fixing non-manifold edges, inverted surface normals, and self-intersecting geometries.
- **Firmware Macro Architecture:** Engineered adaptive bed mesh leveling (`START_PRINT`) and collision-clamped park routines (`PRINT_END`) for Klipper.
- **Slicer Precision Tuning:** Calibrated OrcaSlicer and PrusaSlicer profiles optimizing pressure advance, input shaping, and structural layer bonding.

Autonomous Commerce & Tokenized Fulfillment

`Python (FastAPI)` `Stripe API` `SHA-256` `SQLite`

- **Zero-Trust Tokenized Delivery:** Built cryptographically signed single-use download tokens with 48h expiration and strict download quotas.
- **Idempotent Webhook Engine:** Processed payment event streams with automated signature verification and pre-flight SHA-256 package validation.

✓ VERIFIED TECHNICAL PORTFOLIO • STRICT ON-PLATFORM COMPLIANCE

SPEC / BUILD: 2026.09.21 • REVISION 1.0